

HDOC – 30th day

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Problem Statement:

Write a C program to perform the following menu-driven operations on a given array:

1. Sort the elements using Bubble Sort algorithm and perform a Binary search.
2. Perform Insertion and Deletion in a given valid position.
3. Rotate the array elements to the right by k positions.
4. Exit

Algorithm:

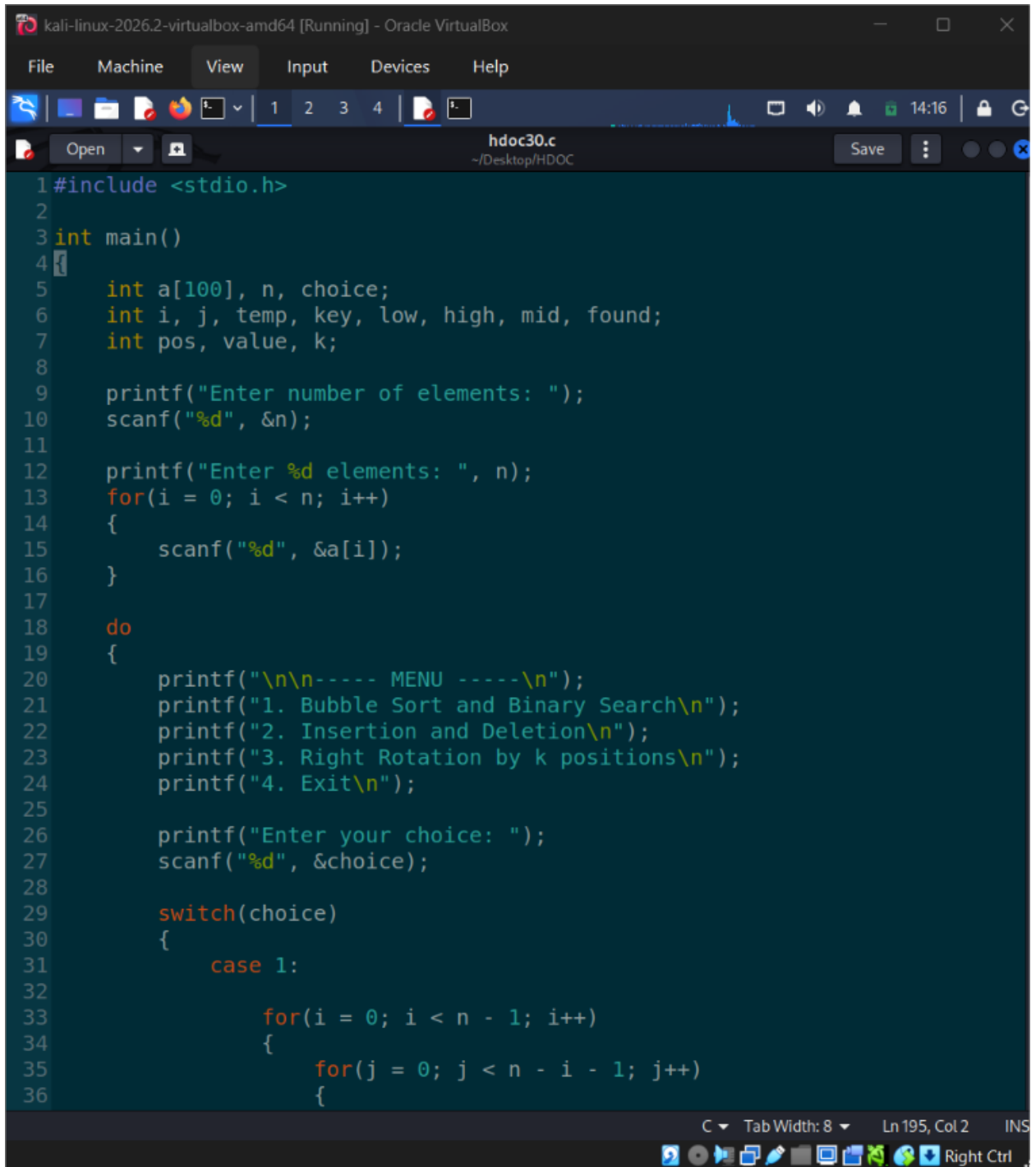
1. Start.
2. Declare an array and required variables.
3. Input the number of elements n.
4. Input n elements into the array.
5. Display the menu:
 - Bubble Sort and Binary Search
 - Insertion and Deletion
 - Right Rotation by k positions
 - Exit
6. Input the user's choice.
7. Use a switch statement to perform the selected operation:
 - Case 1: Sort the array using Bubble Sort. Input the element to search and use Binary Search to find its position.
 - Case 2: Ask whether the user wants insertion or deletion.

- For insertion, input the element and position, shift elements to the right, and insert the element.
 - For deletion, input the position, shift elements to the left, and delete the element.
 - Case 3: Input k. Move the last element to the first position k times to rotate the array to the right.
 - Case 4: Exit the program.
 - Default: Display "Invalid choice".
8. Repeat the menu until the user selects Exit.
9. Stop.

Test Case Table:

Test case	Operation	Input array	Additional input	Expected output	Actual output
1	Bubble sort + Binary search	10 20 30 40 50	Search = 30	Sorted: 10 20 30 40 50, Element found at position 3	Sorted: 10 20 30 40 50, Element found at position 3
2	Insertion	10 20 30 40	Element = 25, Position = 3	10 20 25 30 40	10 20 25 30 40
3	Deletion	10 20 30 40	Position = 3	10 20 40 50	10 20 40 50
4	Right rotation	1 2 3 4 5	K = 2	5 1 2 3 4	5 1 2 3 4
5	Invalid choice	10 20 30 40	Insert at position = 6	Invalid position	Invalid position
6	Exit	Any array	Choice = 4	Program ended	Program ended

C Program:

The image shows a screenshot of a virtual machine window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". Inside the VM, a code editor is open with a file named "hdoc30.c" located at "~/Desktop/HDOC". The code is a C program for a menu-driven sorting and searching application. It includes `<stdio.h>` and defines a `main()` function. The program declares arrays and variables for elements, indices, and choices. It prompts the user to enter the number of elements and then the elements themselves. A menu is displayed with four options: 1. Bubble Sort and Binary Search, 2. Insertion and Deletion, 3. Right Rotation by k positions, and 4. Exit. The program then prompts for a choice and begins to implement the first option, using nested loops for bubble sort. The status bar at the bottom indicates the current tab is "C", the width is 8, and the cursor is at line 195, column 2.

```
1#include <stdio.h>
2
3int main()
4{
5    int a[100], n, choice;
6    int i, j, temp, key, low, high, mid, found;
7    int pos, value, k;
8
9    printf("Enter number of elements: ");
10   scanf("%d", &n);
11
12   printf("Enter %d elements: ", n);
13   for(i = 0; i < n; i++)
14   {
15       scanf("%d", &a[i]);
16   }
17
18   do
19   {
20       printf("\n\n----- MENU ----- \n");
21       printf("1. Bubble Sort and Binary Search\n");
22       printf("2. Insertion and Deletion\n");
23       printf("3. Right Rotation by k positions\n");
24       printf("4. Exit\n");
25
26       printf("Enter your choice: ");
27       scanf("%d", &choice);
28
29       switch(choice)
30       {
31           case 1:
32
33               for(i = 0; i < n - 1; i++)
34               {
35                   for(j = 0; j < n - i - 1; j++)
36                   {
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

hdoc30.c
~/Desktop/HDOC

```
35     for(j = 0; j < n - i - 1; j++)
36     {
37         if(a[j] > a[j + 1])
38         {
39             temp = a[j];
40             a[j] = a[j + 1];
41             a[j + 1] = temp;
42         }
43     }
44 }
45
46 printf("\nSorted Array: ");
47 for(i = 0; i < n; i++)
48 {
49     printf("%d ", a[i]);
50 }
51
52 printf("\nEnter element to search: ");
53 scanf("%d", &key);
54
55 low = 0;
56 high = n - 1;
57 found = 0;
58
59 while(low <= high)
60 {
61     mid = (low + high) / 2;
62
63     if(a[mid] == key)
64     {
65         printf("Element found at position %d\n", mid + 1);
66         found = 1;
67         break;
68     }
69     else if(a[mid] < key)
70     {
```

C Tab Width: 8 Ln 195, Col 2 INS

Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

hdoc30.c
~/Desktop/HDOC

```
69         else if(a[mid] < key)
70         {
71             low = mid + 1;
72         }
73         else
74         {
75             high = mid - 1;
76         }
77     }
78
79     if(found == 0)
80     {
81         printf("Element not found\n");
82     }
83
84     break;
85
86     case 2:
87
88         printf("\n1. Insert");
89         printf("\n2. Delete");
90         printf("\nEnter choice: ");
91         scanf("%d", &j);
92
93         if(j == 1)
94         {
95             printf("Enter element to insert: ");
96             scanf("%d", &value);
97
98             printf("Enter position: ");
99             scanf("%d", &pos);
100
101             if(pos < 1 || pos > n + 1)
102             {
103                 printf("Invalid position\n");
104             }
```

C Tab Width: 8 Ln 195, Col 2 INS
Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc30.c
~/Desktop/HDOC

Save

```
103         printf("Invalid position\n");
104     }
105     else
106     {
107         for(i = n; i >= pos; i--)
108         {
109             a[i] = a[i - 1];
110         }
111
112         a[pos - 1] = value;
113         n++;
114
115         printf("Array after insertion: ");
116
117         for(i = 0; i < n; i++)
118         {
119             printf("%d ", a[i]);
120         }
121     }
122 }
123 else if(j == 2)
124 {
125     printf("Enter position to delete: ");
126     scanf("%d", &pos);
127
128     if(pos < 1 || pos > n)
129     {
130         printf("Invalid position\n");
131     }
132     else
133     {
134         for(i = pos - 1; i < n - 1; i++)
135         {
136             a[i] = a[i + 1];
137         }
138     }
```

C Tab Width: 8 Ln 195, Col 2 INS

Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc30.c
~/Desktop/HDOC

Save

```
133         {
134             for(i = pos - 1; i < n - 1; i++)
135             {
136                 a[i] = a[i + 1];
137             }
138
139             n--;
140
141             printf("Array after deletion: ");
142
143             for(i = 0; i < n; i++)
144             {
145                 printf("%d ", a[i]);
146             }
147         }
148     }
149     else
150     {
151         printf("Invalid choice\n");
152     }
153
154     break;
155
156     case 3:
157
158         printf("Enter value of k: ");
159         scanf("%d", &k);
160
161         k = k % n;
162
163         for(j = 0; j < k; j++)
164         {
165             temp = a[n - 1];
166
167             for(i = n - 1; i > 0; i--)
168             {
169                 a[i] = a[i - 1];
```

C Tab Width: 8 Ln 195, Col 2 INS

Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc30.c
~/Desktop/HDOC

Open Save

```
0
1     k = k % n;
2
3     for(j = 0; j < k; j++)
4     {
5         temp = a[n - 1];
6
7         for(i = n - 1; i > 0; i--)
8         {
9             a[i] = a[i - 1];
10        }
11        a[0] = temp;
12    }
13
14    printf("Array after right rotation: ");
15
16    for(i = 0; i < n; i++)
17    {
18        printf("%d ", a[i]);
19    }
20
21    break;
22
23    case 4:
24        printf("Program ended.\n");
25        break;
26
27    default:
28        printf("Invalid choice!\n");
29    }
30
31    } while(choice != 4);
32
33    return 0;
34
35 }
```

C Tab Width: 8 Ln 195, Col 2

Right Ctrl

Compile the code using:

Gcc hdoc30.c -o hdoc30

```
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc30.c -o hdoc30

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 5
```

Execute the code using:

./hdoc30

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 5
Enter 5 elements: 10 20 30 40 50

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 1

Sorted Array: 10 20 30 40 50
Enter element to search: 30
Element found at position 3

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 5
Enter 5 elements: 10 20 30 40 50

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 2

1. Insert
2. Delete
Enter choice: 1
Enter element to insert: 25
Enter position: 3
Array after insertion: 10 20 25 30 40 50

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.
```

```
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 4
Enter 4 elements: 10 20 30 40

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 2

1. Insert
2. Delete
Enter choice: 2
Enter position to delete: 3
Array after deletion: 10 20 40

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 5
Enter 5 elements: 1 2 3 4 5

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 3
Enter value of k: 2
Array after right rotation: 4 5 1 2 3

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 4
Enter 4 elements: 10 20 30 40

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 2

1. Insert
2. Delete
Enter choice: 1
Enter element to insert: 25
Enter position: 6
Invalid position

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc30
Enter number of elements: 5
Enter 5 elements: 1 2 3 4 5

—— MENU ——
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Right Rotation by k positions
4. Exit
Enter your choice: 4
Program ended.

(kali@kali)-[~/Desktop/HDOC]
$
```